THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET D

Chapters 13 & 14: **Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x + y = 20$, the largest value of xy is: 1

(a) 99 **(b)** 96 **(c)** 120 **(d)** 100

2. In a calendar, the dates in a 2×2 block starting at 5 are 5, 6, 12, 13. Their sum is: 1

(a) 36 **(b)** 28 **(c)** 34 **(d)** 38

3. In a calendar, the dates in a 2×2 block starting at 31 are 31, 32, 38, 39. Their sum is: 1

(a) 132 **(b)** 138 **(c)** 140 **(d)** 142

4. $7 \times 7 - 6 \times 8$ equals: 1

(a) 13 **(b)** 2 **(c)** 1 **(d)** 0

5. If a 3-digit number is written as $\overline{aba}$ its value is: 1

(a) $110a + b$ **(b)** $100a + b$ **(c)** $a + b + a$ **(d)** $101a + 10b$

6. A rectangle has area 150 cm² and length 15 cm. Its breadth is: 1

(a) 135 cm **(b)** 9 cm **(c)** 12 cm **(d)** 10 cm

7. The unit of area is: 1

(a) cubic units **(b)** square units **(c)** units **(d)** none

8. Two triangles on the same base and between the same parallels have: 1

(a) zero area **(b)** equal area **(c)** double area **(d)** different area

9. A square park of side 40 m has area: 1

(a) 800 m² **(b)** 400 m² **(c)** 1600 m² **(d)** 160 m²

10. The area of a rhombus with side 5 cm and one diagonal 6 cm is: 1

(a) 48 cm² **(b)** 12 cm² **(c)** 24 cm² **(d)** 30 cm²

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Expand $(n + 3)^2 - n^2$. 2

12. Write the expression for the sum of the first n odd numbers. 2

13. What is the area of a triangle with vertices forming a right angle with legs 9 and 40? 2

14. Find the area of a rhombus with each side 13 cm and one diagonal 10 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the largest product of two whole numbers with sum 41 and verify against 20 × 21. 3

16. A square of side 14 cm has a smaller square of side 6 cm removed from a corner. Find the remaining area. 3

17. Show that $\overline{ab} \times 11$ gives $a\,(a+b)\,b$ for $a + b < 10$. Check 43×11. 3

OR

Find x in the grid trick: sum of 2×2 block starting at x equals 92. 3

18. A trapezium has area 150 cm², parallel sides 12 cm and 18 cm. Find its height. 3

OR

Find the area of a right triangle with hypotenuse 25 cm and one leg 7 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** In a 3×3 calendar block starting at date x (dates $x, x+1, x+2$ then $x+7$, etc.). 5

(i) Write the centre.

(ii) Show the sum is 9 times the centre.

(iii) Find the sum if the centre is 17.

OR

(i) Why do divisibility tests work for 9 and 11? 5

(ii) Decode: is 8,57,162 divisible by 11?

20. **Case study:** A school ground is a trapezium with parallel sides 120 m and 80 m, and the non-parallel sides 50 m each. 5

(i) Find the height.

(ii) Find the area.

(iii) Find the cost of turf at ₹ 15 per m².

OR

(i) Prove that parallelograms on the same base between the same parallels have equal area. 5

(ii) Illustrate with an example.

END OF QUESTION PAPER — Total: 40 Marks

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